7 Conclusion

Geometry and supports change both the deformation and the linear system. A hole makes the plate softer, holes and inner corners concentrate stress, and moving the load moves the stress peak. Slender bending is the hardest case: linear triangles are too stiff, and the finest beam mesh is 0.7% short of beam theory.

Since A is sparse and symmetric positive definite, sparse LU and CG both apply. We measured $\kappa_2(A)$ growing like N and CG iterations like $N^{1/2}$, as the CG bound predicts, for every object; shape changes only the constant, and the beam is by far the worst conditioned. CG is the best iterative method, diagonal preconditioning gains little, Jacobi diverges and Gauss–Seidel is impractically slow. At these sizes sparse LU is fastest.

Under refinement the displacements settle but peak stresses do not, and the algebraic error of CG ($\leq 10^{-9}$) is far below the mesh changes, so the mesh limits accuracy; supports, force and moment balance and the perturbation bound all check out. Beyond the limitations of Section 6, natural next steps are stronger preconditioners (incomplete Cholesky, multigrid), larger or 3D problems where iterative methods may overtake sparse LU, and higher-order elements.